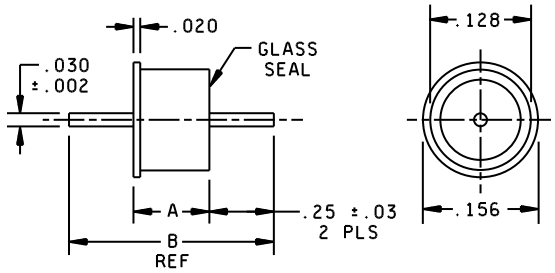
INCH-POUND
MIL-PRF-28861/12H
15 March 2018
SUPERSEDING
MIL-PRF-28861/12G
23 January 2012

PERFORMANCE SPECIFICATION SHEET

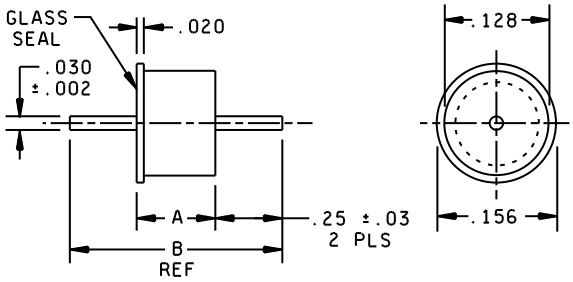
FILTERS, AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE, SUPPERESSION, HERMETICALLY SEALED ON ONE END ONLY, STYLE FS70

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification sheet and MIL-PRF-28861.



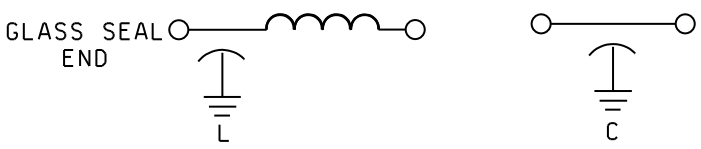
CONFIGURATION A



CONFIGURATION B

FIGURE 1. Case dimensions and circuit diagrams.

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CIRCUIT DIAGRAMS

Dash number	Configuration
001 thru 016 033 and 034	A
017 thru 032 035 and 036	B

Circuit diagram	A ±.005	B Ref
L	.200	.715
C	.110	.625

Inches	mm	Inches	mm
.002	0.05	.156	3.96
.005	0.13	.200	5.08
.020	0.51	.250	6.35
.030	0.76	.625	15.88
.110	2.79	.715	18.16
.128	3.25		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. Circuit diagrams are for information only.
5. Filters shall be supplied with 60/40 solder preform.
6. Potting on non-hermetically sealed end shall not extend beyond .030 inch (0.76 mm) from the filter body.
7. Filters shall be installed using the recommended installation methods (solder-in style of MIL-PRF-28861).

FIGURE 1. Case dimensions and circuit diagrams - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1. Filters and capacitors shall be hermetically sealed on one end as shown in figure 1 for the respective style.

Weight: .25 gram maximum for C circuit parts.
.75 gram maximum for L circuit parts.

Case and lead finish: G only (gold plated).

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: 5 amperes, maximum.

Capacitance: See table I.

Dissipation factor: 2 percent maximum for capacitance values 10 pF through 100pF and 3 percent maximum for values greater than 100pF.

Voltage and temperature limits of capacitance: +15 percent, -40 percent.

Insulation resistance:

At +25°C: 1,000 megohm-microfarads, or 100,000 megohms minimum, whichever is less.

At +125°C: 100 megohm-microfarads or 10,000 megohms minimum, whichever is less.

Insertion loss:

At +25°C: In accordance with table I.

At -55°C and +125°C: A 3 dB degradation from +25°C value shall be allowed.

Voltage drop: 0.05 volt maximum.

DC resistance: 0.01 ohm maximum.

Seal: Not applicable.

Temperature rise: +25°C, maximum.

Solderability of terminals: In accordance with MIL-PRF-28861, except the temperature of the solder shall be +300°C +0°C, -5°C.

Resistance to soldering heat: In accordance with MIL-PRF-28861, except the temperature of the solder shall be +300°C +0°C, -5°C.

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Solderability of mounting termination: In accordance with MIL-PRF-28861, except the temperature of the solder shall be +300°C +0°C, -5°C.

Quality assurance provisions: In accordance with MIL-PRF-28861.

Product assurance level: In accordance with table I.

TABLE I. Electrical Characteristics.

Dash Number	Circuit	Product Assurance level Class		Rated Voltage Volts dc	Capacitance (pF) -0 +100 percent	Minimum insertion loss (dB) in Accordance with MIL-STD-220 1/ 2/					Minimum Insertion loss At resonant Frequency 3/
		B	S			1 Mhz	10 Mhz	100 Mhz	1 Ghz	10 Ghz	
001, 017	L	X	---	50	15,000	7	25	42	50	60	35dB
002, 018	C	X	---	50	15,000	7	25	40	50	60	100Mhz – 1Ghz
003, 019	L	X	X	100	2,700	---	10	25	40	60	--------
004, 020	C	X	X	100	2,700	---	10	25	30	60	
005, 021	L	X	X	100	5,000	---	15	30	35	60	--------
006, 022	C	X	X	100	5,000	---	15	30	35	60	
007, 023	L	X	X	200	10	---	---	---	5	10	--------
008, 024	C	X	X	200	10	---	---	---	4	10	
009, 025	L	X	X	200	25	---	---	---	10	15	5dB
010, 026	C	X	X	200	25	---	---	---	10	15	1Ghz – 10Ghz
011, 027	L	X	X	200	100	---	---	3	20	30	10dB
012, 028	C	X	X	200	100	---	---	3	20	20	1Ghz – 10Ghz
013, 029	L	X	X	200	500	---	---	15	30	50	28dB
014, 030	C	X	X	200	500	---	---	15	30	50	1Ghz – 10Ghz
015, 031	L	X	X	200	1,000	---	4	20	33	55	25dB
016, 032	C	X	X	200	1,000	---	4	20	31	55	1Ghz – 10Ghz
033, 035	L	X	---	50	10,000	4	20	38	50	60	35dB
034, 036	C	X	---	50	10,000	4	20	35	40	60	100Mhz – 1Ghz

1/ For C circuits, insertion loss measurements shall be made under no load. For L circuits, insertion loss measurements shall be made under full load over the frequency range of 1Mhz to 10Mhz. Insertion loss measurements above this frequency range shall be made under no load.

2/ Except as specified in 3/, the insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

3/ The frequency range in which the resonant frequency dip will occur and the minimum insertion loss at the resonant frequency.

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Part or identifying Number (PIN): The PIN shall be as follows:

M28861/12-	XXX	G	X
	Dash no. per figure 1 and table I	Case finish	Product assurance level per MIL-PRF-28861 and table I

Marking: Class B capacitors and filters shall not be marked. Class S filters and capacitors shall be marked with a green dot in accordance with SAE-AMS-STD-595 color ID 14187. Full marking, in accordance with MIL-PRF-28861 shall be marked on the unit package.

Soldering temperature: Caution: These devices shall not be exposed to soldering temperatures exceeding +300°C. Exposure time to soldering temperature of +300°C shall not exceed 1 minute.

Installation note: These devices are intended to be installed into hermetically sealed packages with the glass seal oriented toward the outside world.

Cataloging information: Circuits 'C's shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Circuits 'L's shall be cataloged under FSC 5915 as radio frequency interference filters.

Reference documents: In addition to MIL-PRF-28861, this specification sheet references the following documents.

MIL-STD-220
SAE-AMS-STD-595

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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC
NASA - NA

Preparing activity:
DLA - CC

(Project 59GP-2018-003)

Review activities:
Army - AR, AT, MI
Navy - AS, MC, OS, SH
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